

Response to: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

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Innovation Recommendations

Summary of Key Challenges

- **A multi-faceted AI action plan.** An AI action plan needs to be multi-faceted to be effective. Leadership in AI in the U.S. can be attributed to many features of the U.S. and global scientific enterprise, including
 - o open global dissemination of ideas through scientific publications and computer code sharing.
 - o the dominance of U.S. institutions of higher education in computer science and STEM fields in general, which owes a lot to strong federal funding and their openness and ability to recruit the best talent from across the world;
 - o private funding, a healthy start-up ecosystem, a light regulatory framework, and financial markets that funded the translation of basic research into products and powerful AI models; and
 - o dominance in compute infrastructure that can be attributed to scientific research in chip design and manufacturing and a powerful, government funded compute infrastructure at the various national labs.

For the U.S. to maintain its leading role, particular care is needed to avoid breaking any aspect of the system that led us to a dominant position in AI. The lack of a very stringent regulatory framework can be useful, and it has enhanced all features highlighted above, but, on its own, it cannot prevent the erosion of the U.S. dominant position in AI. For example, curtailing research funding will jeopardize future scientific discoveries and greatly diminish the workforce development that Universities provide for the U.S. to maintain its leadership position in AI.

- **Transparency, explainability, privacy, and accountability.** AI is increasingly affecting and relied upon for many labor- and thought-intensive tasks, from enabling increased automation in manufacturing, automating many business functions and office work, and even performing advanced reasoning for scientific discovery, health care diagnostics, and decision making. Because many decisions made by AI systems affect humans, transparency and accountability is crucial. Incentives and/or mandates are needed to ensure that an AI system can explain/justify a decision or recommendation to any human who is on the receiving end. Similarly, mandates may be needed for companies that offer AI products to disclose processes, what they do with data users contribute (e.g., in AI chatbox interactions), what data have been used for training their models, and what forms of testing their models undertake before they are released to the general public. In general, issues of data provenance, governance, and privacy are critical and so is the ability of users to opt-out of any product features that may infringe on data privacy.
- **Cost of compute and imbalance in computing resources between academia and industry.** Increasingly, there is a large imbalance in the quantity and type of computing resources available at academic institutions compared to the ample resources available in industry. With industry being focused on shorter-term goals, this imbalance can affect future scientific innovation. A robust federally funded computing infrastructure is needed, at the national labs, through equipment grants to academia, and also through efforts like the NAIRR program being piloted by NSF that needs to be maintained and significantly expanded.

Examples of Success and Specific Initiatives to Leverage

1.1 AI Research Infrastructure

- Expand funding for **AI research hubs** leveraging the example of BU's **Hariri Institute for Computing and Computational Science & Engineering** (<https://www.bu.edu/hic/>), which drives interdisciplinary AI research and policy development.
- Strengthen university-industry collaborations through models like BU's **partnership with MassMutual** ([BU-MassMutual](#)), advancing AI-driven financial modeling and risk assessment.
- Support **initiatives like the Massachusetts AI Hub at the Federal level**, integrating research leadership in AI governance and responsible innovation.
- Invest in **high-performance computing resources** at facilities like the [Massachusetts Green High Performance Computing Center \(MGHPCC\)](#), which contributes to scalable AI research and sustainability-focused AI applications.

- Promote open-source AI development and ethical AI frameworks as we do in [BU's AI & Data Science initiatives](#), ensuring broader accessibility and transparency.

1.2 Interdisciplinary AI Applications

- **AI for Public Health & Medicine:** Expand AI-driven disease detection, personalized medicine, and computational drug discovery, leveraging BU's research in biomedical engineering, clinical translation, and public health.
- **AI for Climate & Environmental Science:** Support AI applications in climate modeling, biodiversity tracking, and environmental policy, aligning with [BU's sustainability and urban planning research](#) and [global sustainability](#).
- **AI & Social Sciences:** Invest in AI-driven policy analysis, misinformation tracking, and ethical AI frameworks, integrating law, political science, and communication studies.
- **Creative & Computational AI:** Enhance research on AI-generated art, media studies, and human-AI collaboration, e.g., BU's effort in [digital humanities](#) and computational design.
- **Robotics & Autonomous Systems:** Expand applications in smart transportation, soft robotics, and autonomous decision-making, building on [BU's interdisciplinary robotics initiatives](#)

2. Education Recommendations

BU's AI stance emphasizes the need for all higher education programs to “critically embrace” AI, thereby embracing its transformative potential in every discipline, but also adopting processes that help educate faculty and students in this critical technology, democratizing access, and that seek to understand and mitigate some of the negative impacts. Negative AI impacts include over-reliance, which can lead to atrophy of human skills, sustainability of the compute infrastructure to support AI model training, and ethical implications, such as its potential to be used for harm, accentuating biases, and spreading misinformation. Federal policies need to be informed by such negative impacts, fund research in a restrictive regulatory framework that prevents large excesses and ensures human control of AI systems.

A massive swarth of our future workforce will require generative AI literacy from basic use to its impact on specific fields or applications. The federal government can catalyze a new generation of industry-academic partnerships to ensure that the students graduating from any field have the commensurate AI skills needed to immediately be of value to companies. Examples of this in other domains have recently been described in [Harvard Business Review](#) and other places.



Key Areas of Emphasis Recommended Include:

2.1 AI Literacy Across Disciplines

- Integrate **AI education across all disciplines**, ensuring non-STEM students develop AI literacy.
- Encourage **prompt engineering and GenAI best practices** in higher education curricula.
- Establish **Nationally recognized certification programs** for AI competency.

2.2 Faculty & Student AI Training

- Support **faculty professional development** in AI-driven pedagogy, as we are doing based on the recommendations of BU's AI Task Force.
- Provide support for universities to develop AI-focused courses and workshops.
- Develop centralized repositories of **AI teaching materials**, ensuring equitable access to high-quality learning resources.

2.3 AI Policy & Classroom Implementation

- Define national **guidelines for AI in coursework**.
- Promote **equitable access** to AI-assisted learning tools through partnerships with public and private institutions and Federal initiatives such as NAIRR.
- Fund research into **AI's role in student learning**, emphasizing **ethical AI use and critical engagement**.

Conclusion

These recommendations support a **nationally coordinated AI strategy** that fosters ethical innovation and AI literacy in education. By investing in research infrastructure, interdisciplinary collaboration, and AI education, we can position the U.S. as a global leader in responsible AI development.